

Two Raman phonons measure the fixed edge charge that limits width scaling in monolayer semiconductor nanoribbon transistors

Tanvir M. Mahim^{1, a)} and M. Mosaddequr Rahman¹

Department of Electrical and Electronic Engineering, BRAC University, Dhaka 1212, Bangladesh

(Dated: 29 July 2026)

What limits width scaling in monolayer dichalcogenide nanoribbon transistors is unsettled. We show that two Raman phonons separate edge charge from edge strain. Frozen-phonon calculations on MoS₂, WS₂, MoSe₂ and WSe₂ give a 2LA(M) acoustic overtone about 4.1 times more strain sensitive than the A'_1 mode, whereas only A'_1 renormalises with carrier density. Applied to published tip-enhanced Raman maps, it gives a MoS₂ nanoribbon edge excess of $2.3 \times 10^{12} \pm 7.6 \times 10^{11} \text{ cm}^{-2}$ with strain below 0.03%, while an interior inhomogeneity is pure 0.134% tension. Charge of this size explains a reported depletion-to-enhancement transition on thick oxide but is negligible on a high- κ gate, where the patterning damage halo instead sets a 17 nm critical width; as-grown defect density only rescales the current.

Two-dimensional semiconductors are the leading channel candidates for nanosheet and gate-all-around transistors beyond silicon,^{1–4} and monolayer nanoribbon devices have now been demonstrated at channel widths down to 25 nm, and at current densities above 500 $\mu\text{A}/\mu\text{m}$ in wider ribbons.⁵ As the width shrinks, an increasing fraction of the channel lies within a few nanometres of an etched edge, and the question of what physically stops the scaling becomes pressing. Several candidates have been proposed: geometric edge roughness, conducting edge states,⁶ process-induced damage extending well beyond the patterned line,⁷ and fixed charge trapped at the edge, which was invoked long ago to explain a depletion-to-enhancement transition observed when nanoribbons were trimmed below about 200 nm.⁸ Recent tip-enhanced Raman imaging found a small redshift of the out-of-plane A'_1 phonon at MoS₂ nanoribbon edges and suggested that it was consistent with fixed negative charge, but did not quantify it.⁹ No measured value of the edge charge density in a dichalcogenide nanoribbon exists, and without one the width limit cannot be predicted. Figure 1 shows the device and summarises the argument developed here.

The difficulty is that a phonon frequency responds to strain and to carrier density at once. Etched edges are expected to carry both: elastic relaxation of the free boundary, and charge transfer from dangling bonds and adsorbates. A single mode therefore cannot separate them. The strain-doping correlation plot solves this problem for graphene using two modes with different Grüneisen parameters,¹⁰ and the same idea has been applied to monolayer MoS₂ with the E' and A'_1 pair.¹¹ That pair is poorly conditioned for the present purpose, because both are zone-centre optical phonons with similar and small Grüneisen parameters. Here we use a different pair, the first-order A'_1 mode and the disorder-activated 2LA(M) acoustic overtone, whose strain lever arms differ by a factor of about four. The optical lever arm has been mea-

sured directly under biaxial strain; the acoustic one has not, and we compute it from first principles.

First-principles phonon response. Deformation constants are computed with periodic density functional theory in PySCF,¹² using the PBE functional,¹³ Goedecker-Teter-Hutter pseudopotentials^{14,15} and a DZVP-MOLOPT basis,¹⁶ on the four common 1H monolayers. The A'_1 normal coordinate is the chalcogen half-thickness itself, so its force constant and its relaxed value follow together from the curvature and the minimum of a three-point scan of that coordinate, repeated at three biaxial strains; the in-plane E' mode uses a symmetry-adapted frozen displacement evaluated at both signs, so the force constant comes from a symmetric second difference. A five-point biaxial strain sweep gives the elastic constant, its strain derivative, and the deformation potential of the K -point gap.

Figure 2(a) compares the computed mode Grüneisen parameters with the direct biaxial-strain measurements that exist, which cover both optical modes of MoS₂ and the A'_1 mode of WSe₂.¹⁷ No equivalent biaxial measurement has been published for WS₂ or MoSe₂. The calculated values lie above the measured ones by up to a factor of two for MoS₂, which is the expected sense of a substrate-bending experiment in which only part of the applied strain reaches the monolayer; that same work quantifies the strain transfer loss.¹⁷ Since the separation below depends on the *ratio* of two lever arms, we use the calculated values for both so that they are internally consistent, and show that the result is unchanged if the measured optical value is used instead. As an independent check on the deformation response of the electronic structure, Fig. 2(b) plots the computed strain derivative of the K -point gap against the measured A-exciton gauge factors of the same four materials;¹⁸ the two track each other across the family without any adjustment.

The acoustic branch behaves very differently from the optical modes. Because the zone-edge longitudinal acoustic frequency scales as the square root of the in-plane elastic constant, its Grüneisen parameter follows from the strain dependence of that constant, and we obtain

^{a)} Electronic mail: tanvir.mahim@bracu.ac.bd

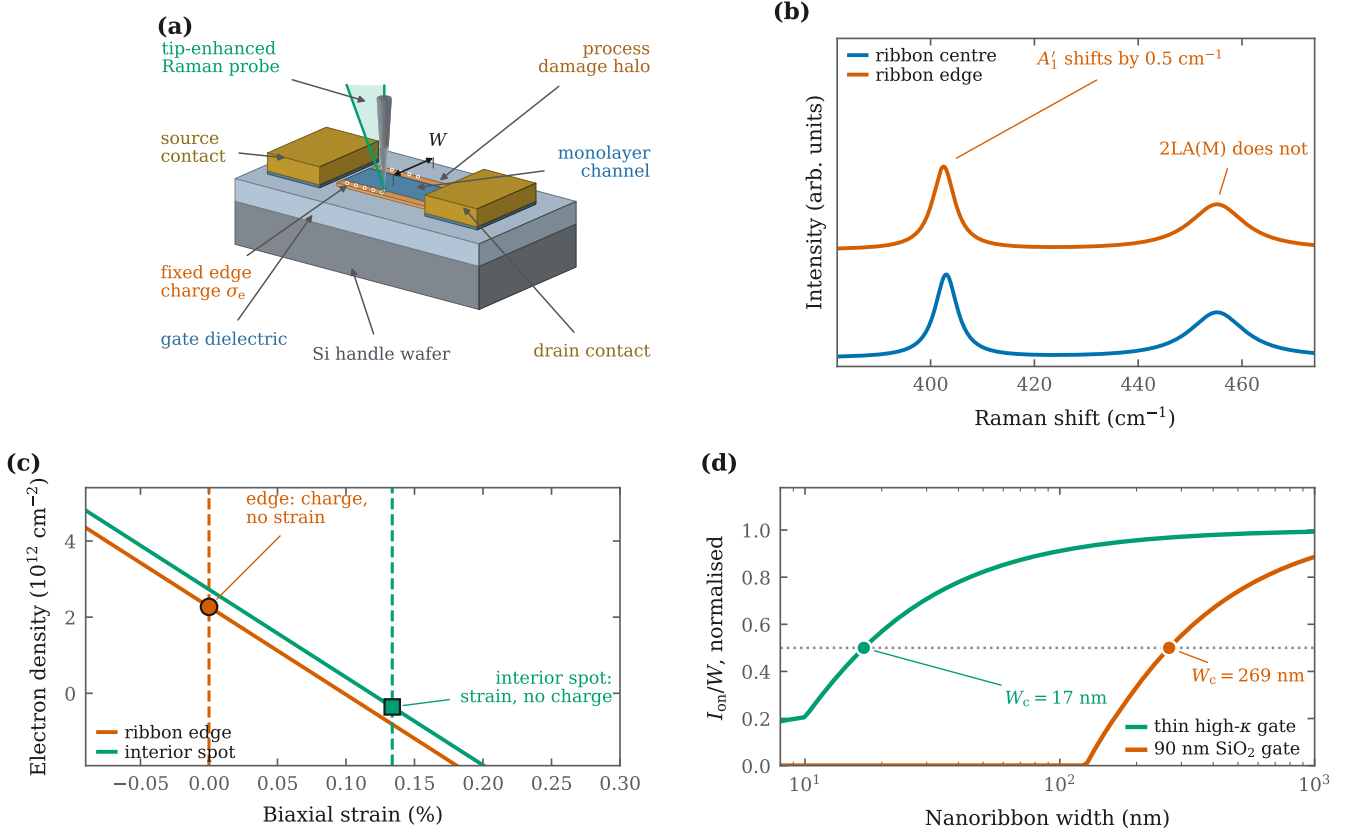


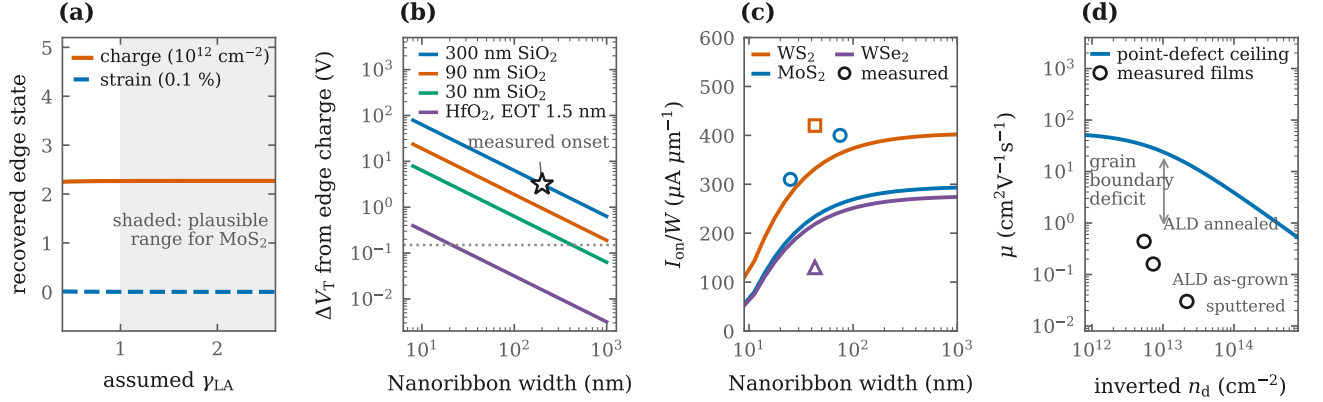
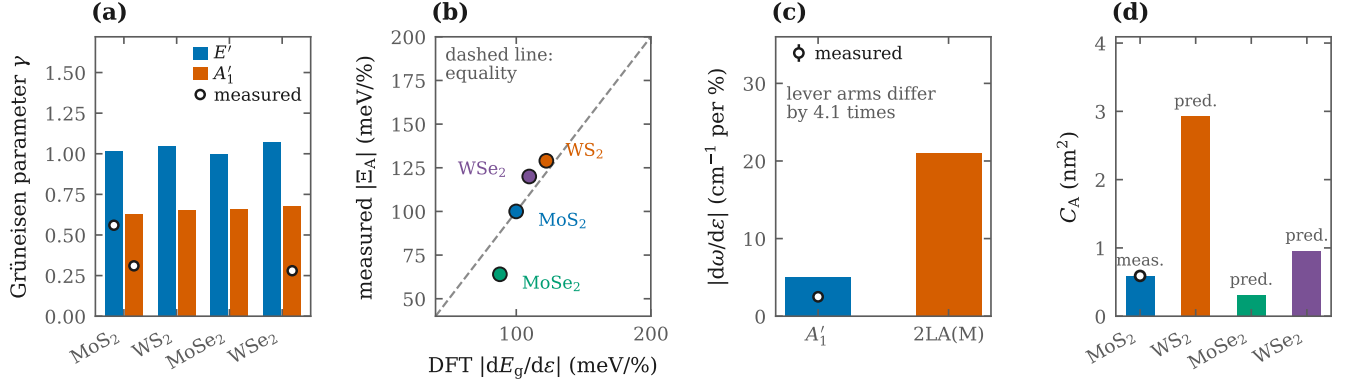
FIG. 1. Fixed edge charge in a monolayer nanoribbon transistor, read out from two Raman phonons. (a) The device: a monolayer ribbon of width W between metal contacts on a gated stack, with the strip of process damage left by patterning and the fixed charge σ_e that accumulates at the etched edges; a tip-enhanced Raman probe samples the ribbon with about ten nanometre resolution. (b) At the edge the A'_1 mode redshifts while the 2LA(M) overtone does not. (c) Each mode gives one constraint in the strain-charge plane: the sloping solid lines are the A'_1 constraint and the vertical dashed lines the 2LA(M) constraint, each in the colour of the location it belongs to. They intersect at finite carrier density and zero strain for the edge, and at finite strain and zero charge for an interior inhomogeneity. (d) The edge charge produces a width-dependent threshold shift, so the width at which the on-current per unit width halves depends strongly on the gate stack. Curves are normalised to the value at $4 \mu\text{m}$, the wide-channel reference used to define W_c , which lies beyond the plotted range.

$\gamma_{\text{LA}} = 2.30$ for MoS_2 . The 2LA(M) overtone at $2\omega_{\text{LA}}$ therefore shifts by 20.9 cm^{-1} per percent of biaxial strain, against 5.1 cm^{-1} per percent for A'_1 , a lever-arm ratio of 4.1 (Fig. 2(c)). This is the property that makes the pair useful: a strain large enough to move A'_1 measurably moves the overtone by several times more. Carrier density acts the other way round. Gated Raman measurements give a softening of A'_1 of 2.2 cm^{-1} per 10^{13} cm^{-2} of electrons, a consequence of the symmetry-selective coupling of the out-of-plane mode to the K -valley electron density.¹⁹ No gated measurement of the 2LA(M) coefficient exists, so we do not assume it vanishes but bound it: allowing the overtone to renormalise with carrier density at up to half the frequency scaled A'_1 rate changes the recovered edge charge by only 16%. The two channels are therefore close to independent.

Edge charge from published nanoscale spectra. Tip-enhanced Raman maps of lithographically patterned monolayer MoS_2 nanoribbons report two distinct

signatures.⁹ At the ribbon edges the A'_1 mode redshifts by 0.5 cm^{-1} while the 2LA(M) overtone does not move. At an interior inhomogeneity of 50 to 100 nm the overtone redshifts by 2.8 cm^{-1} and A'_1 by 0.6 cm^{-1} , with a higher overtone intensity. Solving the two-channel system with the constants above gives, at the edge, $n = 2.3 \times 10^{12} \text{ cm}^{-2}$ of electrons with a strain of $0.000 \pm 0.014\%$, and at the interior spot $\varepsilon = 0.134 \pm 0.014\%$ of tension with no significant charge. Two very different pieces of physics, an electronic one at the boundary and a mechanical one in the interior, are separated by the same measurement.

The edge assignment does not depend on the calculated acoustic Grüneisen parameter. Because the overtone is measured not to shift, the strain is pinned near zero for any value of γ_{LA} , and the whole of the A'_1 shift must be electronic; Fig. 3(a) shows the recovered edge state as γ_{LA} is scanned from 0.4 to 2.6, which brackets the plausible range for MoS_2 . A one-mode analysis could



have assigned the same shift to 0.10% of tension instead, which the overtone excludes because such a strain would have moved 2LA(M) by 2.1 cm^{-1} .

Two caveats attach to the sign and the environment of this number. The coefficient used to convert the frequency shift into a density is measured for free conduction-band electrons, so what the spectrum reports is a local excess of band carriers; in an unbiased sample charge neutrality requires an equal density of ionised edge states to supply them, and it is those states that remain when the device is gated. The sign is set by the environment rather than by the edge itself: the ribbons in question were transferred onto gold, which donates

electrons, whereas nano-optical measurements of MoS₂ edges on oxide report the opposite sense, an edge that is locally depleted.^{20,21} What transfers between environments is the magnitude, of order 10^{12} cm^{-2} within about ten nanometres of the boundary, and it is the magnitude that governs the electrostatics below.

Converting to a line density uses the width over which the excess is distributed. Tip-enhanced Raman resolves about 10 nm, and nano-optical work places the electronic edge region between 5 and 20 nm on oxide,^{6,20} so we take $w_e = 10 \text{ nm}$, giving $\sigma_e = 2.3 \times 10^6 \text{ cm}^{-1}$ and a threshold

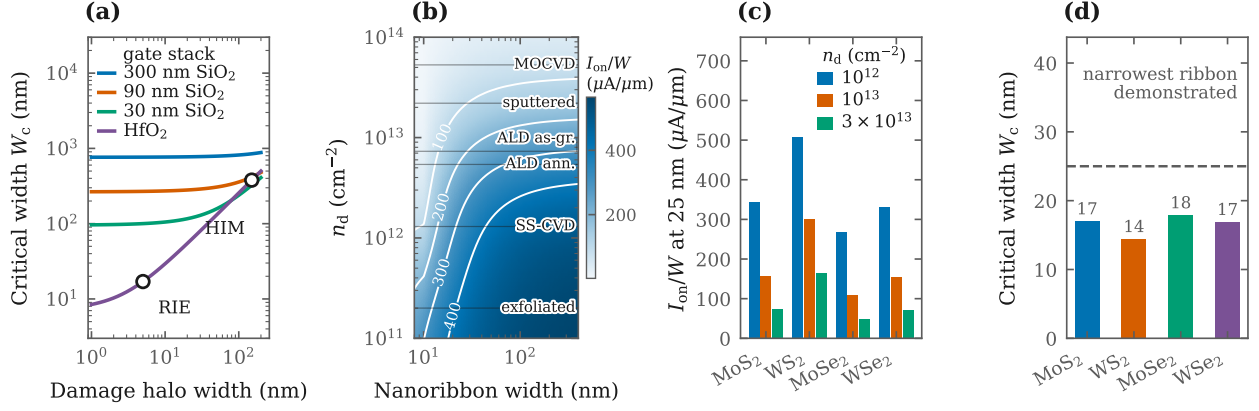


FIG. 4. What sets the width limit. (a) Critical width against process-induced damage halo for four gate stacks; symbols mark reported halo widths for reactive-ion etching (RIE) and helium-ion milling (HIM). (b) On-current per unit width over ribbon width and defect density; horizontal lines mark representative growth technologies, the two atomic-layer-deposited and the sputtered values inverted here from published Raman ratios and the remaining three taken from the literature. (c) Screening of the four materials at a 25 nm width for three defect densities. (d) Critical width of the four materials; the dashed line is the narrowest ribbon demonstrated to date.

shift for a ribbon of width W of

$$\Delta V_T = \frac{2q\sigma_e}{C_{ox}W}. \quad (1)$$

Equation (1) contains no adjustable parameter beyond σ_e and the oxide capacitance, so it can be checked against an experiment performed more than a decade before the spectroscopy. Figure 3(b) shows the predicted shift for four gate stacks. On a 300 nm back-gate oxide it reaches 3.16 V at $W = 200$ nm, comparable to the several-volt threshold movement observed when MoS_2 nanoribbons are trimmed through that width.⁸ On a high- κ stack with an equivalent oxide thickness of 1.5 nm the same charge produces only 0.13 V at $W = 25$ nm, which is why modern nanoribbon devices show no width-driven threshold shift.⁵ Fixed edge charge is not an intrinsic barrier to scaling; it is a barrier only when the gate is weakly coupled.

Carrying the recovered edge charge into a transport model connects it to the measured devices. We combine the phonon-limited mobility with neutral point-defect and screened interface-charge scattering, close the model with the measured monolayer MoS_2 saturation velocity,²² and solve the channel self-consistently: the surface band bending and the electron quasi-Fermi potential are carried as coupled fields, so that the quantum capacitance, the fall of the channel charge towards the drain and the feedback of a series contact resistance all enter without approximation. The coupled system is solved by Newton's method in the open-source device simulator DEVSIM,²³ with the edge charge entering through Eq. (1) and the damage halo through the width-averaged mobility. Taking $V_{ov} = 1.5$ V, $V_{DS} = 1$ V and a 300 nm channel to match the measured devices, this gives 191 and 260 $\mu\text{A}/\mu\text{m}$ for MoS_2 ribbons of width 25 and

75 nm on the high- κ stack, against 310 and 400 $\mu\text{A}/\mu\text{m}$ measured, and 332 $\mu\text{A}/\mu\text{m}$ for WS_2 at 43 nm against 420 $\mu\text{A}/\mu\text{m}$ (Fig. 3(c)).⁵ The model therefore reproduces the measured width dependence and the ordering of the materials while sitting between 21 and 38% below the absolute level, which is what a semiclassical model carrying no fitted transport parameter can be expected to do.

The p-type WSe_2 device is the informative exception. The channel model predicts 218 $\mu\text{A}/\mu\text{m}$ against 130 $\mu\text{A}/\mu\text{m}$ measured, and reproducing the measurement requires a contact resistance of 2.7 $\text{k}\Omega \mu\text{m}$ per contact, about 5 times the 560 $\Omega \mu\text{m}$ reported for the n-type devices of the same work.⁵ That is the known difficulty of making a p-type contact to WSe_2 ,^{24,25} and the self-consistent solve separates it from the channel rather than leaving it as an unquantified caveat.

What sets the width limit. The second control on scaling is the damage halo left by patterning, a strip beside each edge with an elevated defect density. Reported halo widths span two orders of magnitude, from a few nanometres for optimised reactive-ion etching⁵ to about 150 nm for helium-ion milling.⁷ Defining the critical width W_c as the width at which the on-current per unit width falls to half of its wide-channel value at $V_{ov} = 1.5$ V, Fig. 4(a) plots W_c against halo width for the four gate stacks. The two causes act in different regimes and are worth separating. On a thin high- κ gate with a narrow halo the halo dominates, contributing 14 nm against 7 nm from the edge charge, and $W_c = 17$ nm; the demonstrated 25 nm ribbons sit just above it. For a wide halo the critical width tracks the halo directly and reaches 379 nm. On a thick oxide the edge-charge term takes over and W_c saturates near 96 nm for a 30 nm oxide, no matter how clean the etch is. The distinction matters in practice, because a width-dependent threshold offset can be partly com-

pensated by design, whereas the transport degradation of the damaged halo cannot. The edge-charge branch scales with the assumed edge width, moving W_c on the high- κ stack from 15 to 21 nm as w_e is varied from 5 to 20 nm, whereas the halo branch is unaffected.

As-grown film quality enters differently. To make that explicit we invert the disorder-activated Raman ratio of published atomic-layer-deposited and sputtered MoS₂ films,²⁶ using the relation $I_{LA}/I_{A'_1} = C_A n_d$ calibrated on ion-irradiated monolayers.²⁷ Those films span n_d from 5.4×10^{12} to 2.2×10^{13} cm⁻², and their measured mobilities lie between 74 and 479 times below the point-defect ceiling (Fig. 3(d)), the expected signature of nanocrystalline material in which grain boundaries rather than point defects dominate transport.²⁸ Over that same range of defect density the on-current at a fixed 25 nm width falls by a factor of 2.3 (Fig. 4(b) and (c)), while W_c moves by only 6%. Film quality therefore sets how much current a ribbon delivers, whereas the patterning process and the gate stack set how narrow it can usefully be made.

Because the calibration constant C_A has been measured only for MoS₂, we also predict it for the rest of the family. In the double resonance picture the disorder-activated intensity relative to the first-order line scales as $|D|^2/(\omega_{LA}^2 \Delta^2)$, with D the gap deformation potential computed above, ω_{LA} the zone-edge acoustic frequency and Δ the detuning of the laser from the A exciton. Anchoring on the MoS₂ value 0.59 ± 0.03 nm² at 532 nm,²⁷ this gives $C_A = 2.92, 0.31$ and 0.95 nm² for WS₂, MoSe₂ and WSe₂ (Fig. 2(d)), testable predictions that would let the same ion-irradiation calibration be skipped for those materials. Screening the family at a 25 nm width puts WS₂ first on current density and also gives it the smallest critical width, so it is the most scalable of the four on both counts (Fig. 4(c) and (d)); the ordering follows the effective masses and phonon-limited mobilities.^{29,30}

Both separations above are two-observable weighted least squares with a broad prior. The same forward model extends to a full hyperspectral image, where defect density, strain and carrier density are recovered jointly; that extension uses a physics-guided adjoint solver and is verified on synthetic data in the supplementary material, together with the complete parameter provenance. The code is openly available.

Several limitations bound these conclusions. The electronic structure is computed at the PBE level without spin-orbit coupling, so we use published spin-orbit-resolved effective masses^{29,31} and take only strain derivatives from our own calculations. The edge charge is inferred from a single published data set on one material, on ribbons supported by gold, and assumes the excess is uniform over an edge width of 10 nm; the sign is environment dependent and the reported density scales inversely with the measured A'_1 doping coefficient. A capacitance measurement on a width series would test all three. The predicted calibration constants for WS₂, MoSe₂ and WSe₂ rest on a double-resonance scaling argument and await measurement. The transport layer contains two

constants fixed once against published results, the neutral defect potential and the interface charge density of each gate stack. Currents quoted against measurement come from a self-consistent Poisson and drift-diffusion solve that includes contact resistance. The design map and the critical-width scans use the faster compact expression, which overestimates the absolute current by a nearly constant factor, the self-consistent to compact ratio being 0.64; the critical width, being a ratio of currents, differs between the two models by only 7%. The solve is semiclassical, so quantum transport is not treated and short-channel electrostatics enter only through the assumed channel length.

In summary, the quantity that limits width scaling in monolayer dichalcogenide nanoribbon transistors is fixed charge at the etched edge, and it can be read out non-destructively from the ratio of two phonon responses. First-principles Grüneisen parameters make the A'_1 and 2LA(M) pair an orthogonal probe of charge and strain, and applying it to published tip-enhanced Raman data yields a local excess of 2.3×10^{12} cm⁻² of band electrons at MoS₂ nanoribbon edges, and therefore an equal density of fixed ionised edge states, together with a clean separation of a purely mechanical interior inhomogeneity. Charge of that size accounts for a decade-old depletion-to-enhancement observation and, together with the patterning damage halo, sets a critical width ranging from 17 nm on a thin high- κ gate to several hundred nanometres for a weak gate or a damaging etch, while depending only weakly on as-grown defect density. The practical consequence is that width scaling is bought with gate coupling and gentle patterning rather than with better films, and that both can be checked optically before a device is built.

AUTHOR DECLARATIONS

Conflict of interest. The authors have no conflicts to disclose.

Author contributions. **Tanvir M. Mahim:** Conceptualization (lead); Methodology (lead); Software (lead); Formal analysis (lead); Writing – original draft (lead). **M. Mosaddequr Rahman:** Supervision (lead); Validation (equal); Writing – review and editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.21670316>, reference number 32. That deposit contains the raw first-principles output, every digitised published measurement used here as a flat file with its source stated, the machine-readable results file from which every number, table and figure in this article is generated, and the figures as published. The complete code base, including the density functional theory drivers, the

forward model, the adjoint solver, the transport layer and the scripts that regenerate every figure and every number in this article, is openly available at <https://github.com/Tanvir-Mahmud-Mahim/Width-scaling-in-monolayer-semiconductor-nanoribbon-transistors> and is archived in the same Zenodo deposit.

- ¹A. Pal, T. Chavan, J. Jabbour, W. Cao, and K. Banerjee, “Three-dimensional transistors with two-dimensional semiconductors for future CMOS scaling,” *Nat. Electron.* **7**, 1147–1157 (2024).
- ²L. Yin, R. Cheng, J. Ding, J. Jiang, Y. Hou, X. Feng, Y. Wen, and J. He, “Two-dimensional semiconductors and transistors for future integrated circuits,” *ACS Nano* **18**, 7739–7768 (2024).
- ³J. Tang, J. Jiang, X. Gao, X. Gao, C. Zhang, M. Wang, C. Xue, Z. Li, Y. Yin, C. Tan, F. Ding, C. Qiu, L.-M. Peng, and H. Peng, “Low-power 2D gate-all-around logics via epitaxial monolithic 3D integration,” *Nat. Mater.* **24**, 519–526 (2025).
- ⁴Y. H. Kim, D. Lee, W. Huh, J. Lee, D. Lee, G. Wang, J. Park, D. Ha, and C.-H. Lee, “Gate stack engineering of two-dimensional transistors,” *Nat. Electron.* **8**, 770–783 (2025).
- ⁵T. Peña, A. E. O. Persson, A. Krayev, Á. Fridriksdóttir, H. Su, Y.-M. Lee, Y. S. Song, K. Neilson, Z. Zhang, A. T. Hoang, J. A. Yang, L. Hoang, S. X. Wang, A. J. Mannix, P. C. McIntyre, and E. Pop, “Scaling nanoribbon transistors with monolayer transition metal dichalcogenides,” *Nat. Nanotechnol.* **21**, 803–809 (2026).
- ⁶D. Wu, X. Li, L. Luan, X. Wu, W. Li, M. N. Yogeesh, R. Ghosh, Z. Chu, D. Akinwande, Q. Niu, and K. Lai, “Uncovering edge states and electrical inhomogeneity in MoS₂ field-effect transistors,” *Proc. Natl. Acad. Sci. U.S.A.* **113**, 8583–8588 (2016).
- ⁷X. Liu, D. Zemlyanov, S. Sharma, S. Ghosh, J. Lasseter, T. E. Beechem, J. Appenzeller, S. J. Randolph, and Z. Chen, “Focused helium ion beam for direct patterning of monolayer MoS₂ nanoribbon field effect devices,” *Adv. Funct. Mater.* **36**, 2514880 (2026).
- ⁸H. Liu, J. Gu, and P. D. Ye, “MoS₂ nanoribbon transistors: transition from depletion mode to enhancement mode by channel-width trimming,” *IEEE Electron Device Lett.* **33**, 1273–1275 (2012).
- ⁹A. Krayev, T. Peña, A. E. O. Persson, K. Neilson, A. T. Hoang, A. J. Mannix, and E. Pop, “Nano-Raman imaging of monolayer MoS₂ nanoribbons,” *Appl. Phys. Lett.* **128**, 203102 (2026).
- ¹⁰J. E. Lee, G. Ahn, J. Shim, Y. S. Lee, and S. Ryu, “Optical separation of mechanical strain from charge doping in graphene,” *Nat. Commun.* **3**, 1024 (2012).
- ¹¹A. Michail, N. Delikoukos, J. Parthenios, C. Galiotis, and K. Papagelis, “Optical detection of strain and doping inhomogeneities in single layer MoS₂,” *Appl. Phys. Lett.* **108**, 173102 (2016).
- ¹²Q. Sun, X. Zhang, S. Banerjee, P. Bao, M. Barbry, N. S. Blunt, N. A. Bogdanov, G. H. Booth, J. Chen, Z.-H. Cui, *et al.*, “Recent developments in the PySCF program package,” *J. Chem. Phys.* **153**, 024109 (2020).
- ¹³J. P. Perdew, K. Burke, and M. Ernzerhof, “Generalized gradient approximation made simple,” *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
- ¹⁴S. Goedecker, M. Teter, and J. Hutter, “Separable dual-space Gaussian pseudopotentials,” *Phys. Rev. B* **54**, 1703–1710 (1996).
- ¹⁵C. Hartwigsen, S. Goedecker, and J. Hutter, “Relativistic separable dual-space Gaussian pseudopotentials from H to Rn,” *Phys. Rev. B* **58**, 3641–3662 (1998).
- ¹⁶J. VandeVondele and J. Hutter, “Gaussian basis sets for accurate calculations on molecular systems in gas and condensed phases,” *J. Chem. Phys.* **127**, 114105 (2007).
- ¹⁷A. Michail, G. Anagnostopoulos, S. Katsiaounis, J. Parthenios, and K. Papagelis, “Biaxial strain transfer in monolayer MoS₂ and WSe₂ transistor structures,” *ACS Appl. Mater. Interfaces* **16**, 49602–49611 (2024).
- ¹⁸E. Henríquez-Guerra, H. Li, M. Serrano-Ruiz, M. Peralta, S. Reyes-Vega, G. Pizzochera, P. Ares, F. Zamora, A. Castellanos-Gomez, M. D. Jimenez-Sanchez, D. Costa-Riquelme, C. Munuera, L. Vaquero-Garzon, J. J. Baldoví, R. Frisenda, and M. L. Ruíz-González, “Large biaxial compressive strain tuning of neutral and charged excitons in single-layer transition metal dichalcogenides,” *ACS Appl. Mater. Interfaces* **15**, 57369–57378 (2023).
- ¹⁹B. Chakraborty, A. Bera, D. V. S. Muthu, S. Bhowmick, U. V. Waghmare, and A. K. Sood, “Symmetry-dependent phonon renormalization in monolayer MoS₂ transistor,” *Phys. Rev. B* **85**, 161403 (2012).
- ²⁰W. Su, N. Kumar, S. J. Spencer, N. Dai, and D. Roy, “Optical probing of the edge-related properties of atomically thin MoS₂ at the nanoscale,” *Nat. Commun.* **10**, 5544 (2019).
- ²¹F. B. Sousa, R. Nadas, R. Martins, A. P. M. Barboza, J. S. Soares, B. R. A. Neves, I. Silvestre, A. Jorio, and L. M. Malard, “Disentangling doping and strain effects at defects of grown MoS₂ monolayers with nano-optical spectroscopy,” *Nanoscale* **16**, 12923–12933 (2024).
- ²²K. K. H. Smithe, C. D. English, S. V. Suryavanshi, and E. Pop, “High-field transport and velocity saturation in synthetic monolayer MoS₂,” *Nano Lett.* **18**, 4516 (2018).
- ²³J. E. Sanchez, “DEVSIM: A TCAD semiconductor device simulator,” *J. Open Source Softw.* **7**, 3898 (2022).
- ²⁴Z. Sun, A. Afzal, P. Wu, H. Zhang, S. Krylyuk, R. Tripathi, H.-Y. Lan, Y. Tan, J. Cai, G. Pourtois, A. Davydov, Z. Chen, and J. Appenzeller, “Low-resistance contacts for *p*-type monolayer tungsten diselenide transistors using metallic layered Nb_{0.3}W_{0.7}Se₂,” *Nat. Electron.* **9**, 358–366 (2026).
- ²⁵A. Mondal, C. Biswas, S. Park, W. Cha, S.-H. Kang, M. Yoon, S. H. Choi, K. K. Kim, and Y. H. Lee, “Low Ohmic contact resistance and high on/off ratio in transition metal dichalcogenides field-effect transistors via residue-free transfer,” *Nat. Nanotechnol.* **19**, 34–43 (2024).
- ²⁶C. A. Nattoo, T. Peña, K. N. Nazif, X. Wu, S. Rahimisheikh, G. D’Acunto, S. F. Bent, J. Hadermann, and E. Pop, “Optoelectronic properties of atomic layer deposited and sputtered MoS₂ films,” *ACS Appl. Mater. Interfaces* **17**, 47347–47354 (2025).
- ²⁷S. Mignuzzi, A. J. Pollard, N. Bonini, B. Brennan, I. S. Gilmore, M. A. Pimenta, D. Richards, and D. Roy, “Effect of disorder on Raman scattering of single-layer MoS₂,” *Phys. Rev. B* **91**, 195411 (2015).
- ²⁸A. M. van der Zande, P. Y. Huang, D. A. Chenet, T. C. Berkelbach, Y. You, G.-H. Lee, T. F. Heinz, D. R. Reichman, D. A. Muller, and J. C. Hone, “Grains and grain boundaries in highly crystalline monolayer molybdenum disulphide,” *Nat. Mater.* **12**, 554–561 (2013).
- ²⁹Z. Jin, X. Li, J. T. Mullen, and K. W. Kim, “Intrinsic transport properties of electrons and holes in monolayer transition-metal dichalcogenides,” *Phys. Rev. B* **90**, 045422 (2014).
- ³⁰V.-A. Ha and F. Giustino, “High-throughput screening of 2D materials identifies *p*-type monolayer WS₂ as potential ultra-high mobility semiconductor,” *npj Comput. Mater.* **10**, 229 (2024).
- ³¹A. Kormányos, G. Burkard, M. Gmitra, J. Fabian, V. Zólyomi, N. D. Drummond, and V. Fal’ko, “*k*·*p* theory for two-dimensional transition metal dichalcogenide semiconductors,” *2D Mater.* **2**, 022001 (2015).
- ³²T. M. Mahim and M. M. Rahman, “Two Raman phonons measure the fixed edge charge that limits width scaling in monolayer semiconductor nanoribbon transistors: data, benchmarks and code,” (2026), zenodo deposit.